Evanescence Reaction in Subdiffusion Molecular Communication Channel

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ABSTRACT

This work considers the subdiffusion motion of information carrying molecules inside a molecular communication (MC) channel. Additionally, an evanescence process is taken into account, which leads to the degradation of molecules prior to their hitting to the receiver's boundary. The closed-form expressions of the arrival probability and the first-passage-time-density (FPTD) are obtained. Furthermore, the performance of MC was investigated by using the concentration-based modulation technique. The probability of detection is presented for different parameters of a reaction rate. The performance results were validated by the Monte-Carlo simulation.

CCS CONCEPTS

• **Networks** → **Network performance analysis**; **Network simulations**.

KEYWORDS

molecular communication, subdiffusion, arrival probability

ACM Reference Format:
Nadezhda Briantceva, Lokendra Chouhan, Matteo Parsani, and Mohamed-Slim Alouini. 2022. Evanescence Reaction in Subdiffusion Molecular Communication Channel. In *The Ninth Annual ACM International Conference on Nanoscale Computing and Communication (NANOCOM '22), October 5–7, 2022, Barcelona, Spain.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3558583.3558868

1 INTRODUCTION

Molecular communication (MC) is a rapidly developing research field where information exchange happens by chemical means. In this work, we consider subdiffusion MC channel, which can be found in complex (porous or fractal) and crowded environments (molecule's motion in living cells) where the molecule's mean square displacement scales as a fractional order power law in time. Target search processes driven by subdiffusion motion, where a reaction or trapping event occurs after meeting two objects, have attracted considerable interest in recent years [2]. In Fig. 1, we can find the schematic representation of the considered in this work 3D system.

NANOCOM '22, October 5–7, 2022, Barcelona, Spain

ACM ISBN 978-1-4503-9867-1/22/10.
https://doi.org/10.1145/3558583.3558868

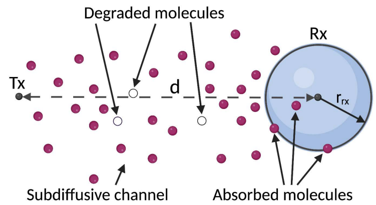

Figure 1: Subdiffusion-reaction system model.

2 MAIN RESULTS

In our work, we combine the effects of the reaction and the subdiffusion to be able to study the information molecule's survival probability. It can be expressed through the probability density $S_1^*(r_0, t; R) = \int c(r, t \mid r_0; 0)\, dr$ and, thus, it obeys [1]:

$$\frac{\partial S_1^*(r,t;R)}{\partial t} = \frac{\alpha(t)}{\alpha_0} D_{\gamma 0} D_t^{1-\gamma} \left(\frac{\alpha_0}{\alpha(t)} \nabla_{\mathrm{r}}^2 \right) S_1^*(r,t;R) + \frac{\dot{\alpha}(t)}{\alpha(t)} S_1^*(r,t;R), \tag{1}$$

with the following initial and boundary conditions:

$$S_1^*(r,0;R) = 1, S_1^*(R,t;R) = 0, \lim_{r\to\infty} S_1^*(r,t;R) = \frac{\alpha(t)}{\alpha_0}. \tag{2}$$

The survival probability can be derived from knowing the solution of the problem without evanescence [1], [4]. Thus, $S_1^*(r,t;R) = \frac{\alpha(t)}{\alpha_0} S_1(r,t;R)$, where $S_1(r,t;R)$ is the non-evanescent survival probability. In our work, we consider an exponentially decaying molecule density, $\alpha(t) = \alpha_0 e^{(-\lambda t)}$, and, therefore, the solution of the problem (1)-(2) can be expressed as:

$$S_1^*(r,t;R) = e^{-\lambda t} \left(1 - \left(\frac{R}{r}\right)^{\alpha} H_{11}^{10} \left[\frac{r-R}{\sqrt{D_\gamma t^\gamma}} \;\middle|\; \begin{matrix} (1, \gamma/2) \\ (0,1) \end{matrix} \right] \right). \tag{3}$$

Using the above expression, the arrival probability $B_1^*(r,t;R) = 1 - S_1^*(r,t;R)$ can be written as:

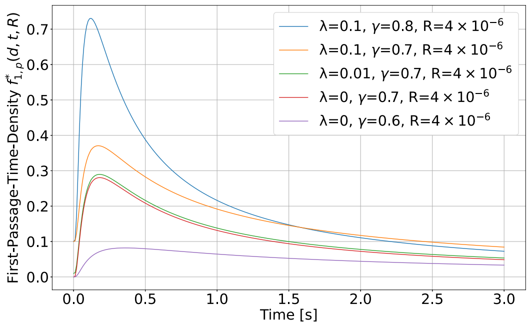

Figure 2: Information molecule FPTD with different reaction rate parameters in 1D.

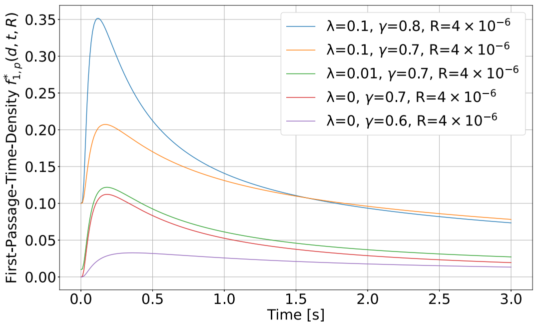

Figure 3: Information molecule FPTD with different reaction rate parameters in 3D.

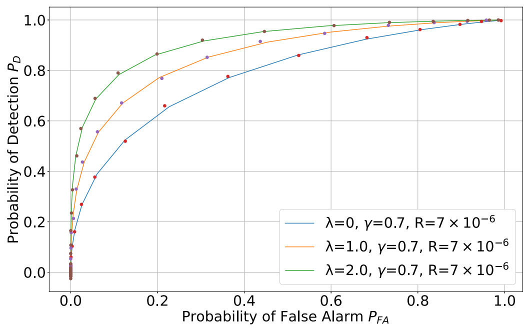

Figure 4: Probability of detection versus probability of false alarm in 1D.

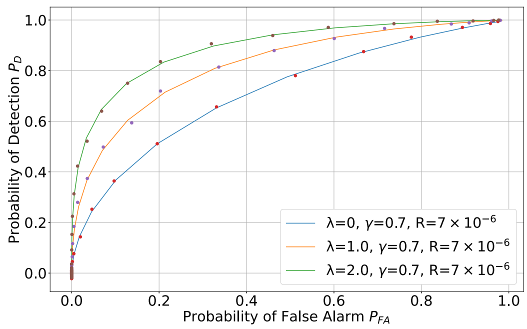

Figure 5: Probability of detection versus probability of false alarm in 3D.

$$B_1^*(r,t;R) = 1 - e^{-\lambda t}\left(1 - \left(\frac{R}{r}\right)^{\alpha} H_{11}^{10}\left[\frac{r-R}{\sqrt{D_\gamma t \gamma}} \;\middle|\; \begin{matrix}(1,\gamma/2)\\(0,1)\end{matrix}\right]\right). \tag{4}$$

Moreover, from the FPTD expression of a molecule $f^*_{1,\mathrm{p}}(r,t) = -\frac{\partial}{\partial t}S_1^*(r,t;R)$, we get:

$$f^*_{1,p}(r,t;R) = e^{-\lambda t}\left(\lambda - \left(\frac{R}{r}\right)^{\alpha}\left(\lambda H_{11}^{10}\left[\frac{r-R}{\sqrt{D_\gamma t^\gamma}} \;\middle|\; \begin{matrix}(1,\gamma/2)\\(0,1)\end{matrix}\right] - \frac{\gamma}{2t}H_{11}^{10}\left[\frac{r-R}{\sqrt{D_\gamma t^\gamma}} \;\middle|\; \begin{matrix}(1,\gamma/2)\\(1,1)\end{matrix}\right]\right)\right). \tag{5}$$

In Fig. 2 and 3, we present the analytical results of the FPTD for both 1D and 3D cases. We can see that the results closely match but with higher FPTD peaks in the 1D case. Moreover, it can be noticed that by taking a lower reaction rate λ and fractional power γ, the peak values of FPTD decrease.

Next, we analyze the impact of the evanescence reaction on the performance of a subdiffusive MC system. Here we use the same type of molecules for the information transfer [3]. In Fig. 4 and 5, the results of the probability of false alarm along with the probability of detection are presented. The obtained results were verified by the Monte-Carlo simulation [5]. It can be seen from the plots how performance is getting better with increasing the reaction rate values. Indeed, this is due to the degradation reaction, which leads to the molecule's absorption and further reduction of the inter-symbol interference (ISI).

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